

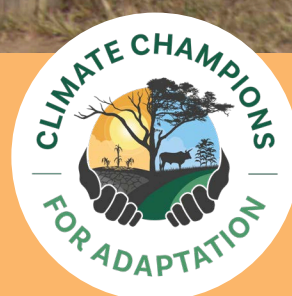
#1: AN INTRODUCTION TO CLIMATE CHANGE

**PUBLICATION SERIES ON UNDERSTANDING CLIMATE CHANGE
AND ITS IMPACTS ON SMALL-SCALE FARMERS IN SOUTH AFRICA**

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WHAT IS CLIMATE CHANGE?

Climate change refers to long-term changes in the usual weather patterns we experience in a region, or over the Earth as a whole. These changes can include, for example, how much rain typically falls in a year, or changes in the usual monthly or seasonal temperatures in any specific location. While the weather we experience can change in a few hours, climate generally takes hundreds or even millions of years to change.

We started noticing significant changes to the planet's climate from the middle of the 20th Century^[1]. These are being caused by the increased levels of greenhouse gasses in the atmosphere. The most significant of these gasses is carbon dioxide, which is largely produced by people burning organic material, especially fossil fuels such as coal, oil and natural gas. Industrial processes, transport systems, deforestation and agriculture all contribute to the increase of greenhouse gasses in the atmosphere. These gases trap the heat from the sun's rays inside the earth's atmosphere, causing the temperature to rise.

While the earth's climate does change naturally over time, this process usually takes place very slowly. Current global warming and the other changes to the climate that are being driven by this increased heat are taking place at a much faster pace than we or the natural environment can adapt to. We call this anthropogenic (man-made) climate change.

THE GREENHOUSE EFFECT AND GREENHOUSE GASES

To understand climate change we need to understand how the greenhouse effect works. The greenhouse effect is a natural process whereby the gasses in the Earth's atmosphere trap some of the heat from the sun, warming the Earth and making it liveable. During the day, heat from the sun warms the earth's surface. Some of this heat is radiated back into space, but the greenhouse gasses in the atmosphere trap some of this heat, keeping the earth consistently warm.

Human activities since the start of the Industrial Revolution have been changing the earth's natural

greenhouse effect by releasing far higher quantities of these greenhouse gases. By burning fossil fuels, and by deforesting and cultivating the land, we have released vast amounts of carbon that had previously been safely stored in the earth's crust and in the soil. This carbon is released into the atmosphere in the form of carbon dioxide. This is enhancing the greenhouse effect and warming the earth far faster than would naturally occur, causing climate change.

The major greenhouse gases responsible for causing climate change are carbon dioxide (CO₂), methane (CH₄) and nitrous oxide (N₂O). Some man-made fluorinated gases are less widespread, but are nevertheless very powerful greenhouse gases, for example hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), and sulphur hexafluoride (SF₆). Greenhouse gases vary in how well they absorb energy and how long they stay in the atmosphere, which combine to make their Global Warming Potential (GWP). To make it easy to compare greenhouse gases we usually compare them to the most common greenhouse gas – carbon dioxide – by calculating their carbon dioxide equivalent (CO₂-eq).



Letabo Coal Power Station, Free State Province, South Africa. (Source: T Lebert)

Cover photo: Woman farmer, KwaZulu-Natal, Heifer Project Southern Africa (HPSA)

^[1] Intergovernmental Panel on Climate Change (IPCC), 2021

THE EFFECTS OF CLIMATE CHANGE

Even small changes in the Earth's temperature can have major impacts on our climate system. The Earth's temperature has already increased by an average of $1.1^{\circ}\text{C}^{[2]}$ in the past century and scientists agree that global temperatures are likely to keep increasing over the next 100 years. Climate impacts span the physical environment, ecosystems and human societies.

Climatologists have looked at the evidence from historical sources such as ice cores, tree rings, and layers of ocean and rock sediment to understand past patterns and changes in the Earth's climate^[3]. This tells us that the current warming is happening roughly ten times faster than the average rate of warming following an ice age, while carbon dioxide levels are increasing more than 250 times faster than after the last ice age.

Many of the physical impacts of climate change are already visible and have been confirmed in the internationally recognised reports of the Intergovernmental Panel on Climate Change (IPCC). Warming temperatures are causing ice sheets and glaciers to melt, while also causing ocean waters to expand, resulting in sea level rise which damages and floods low-lying coastal areas.

Climate change is causing weather patterns to be less predictable and is linked to more frequent damaging weather events such as droughts, floods, and tropical

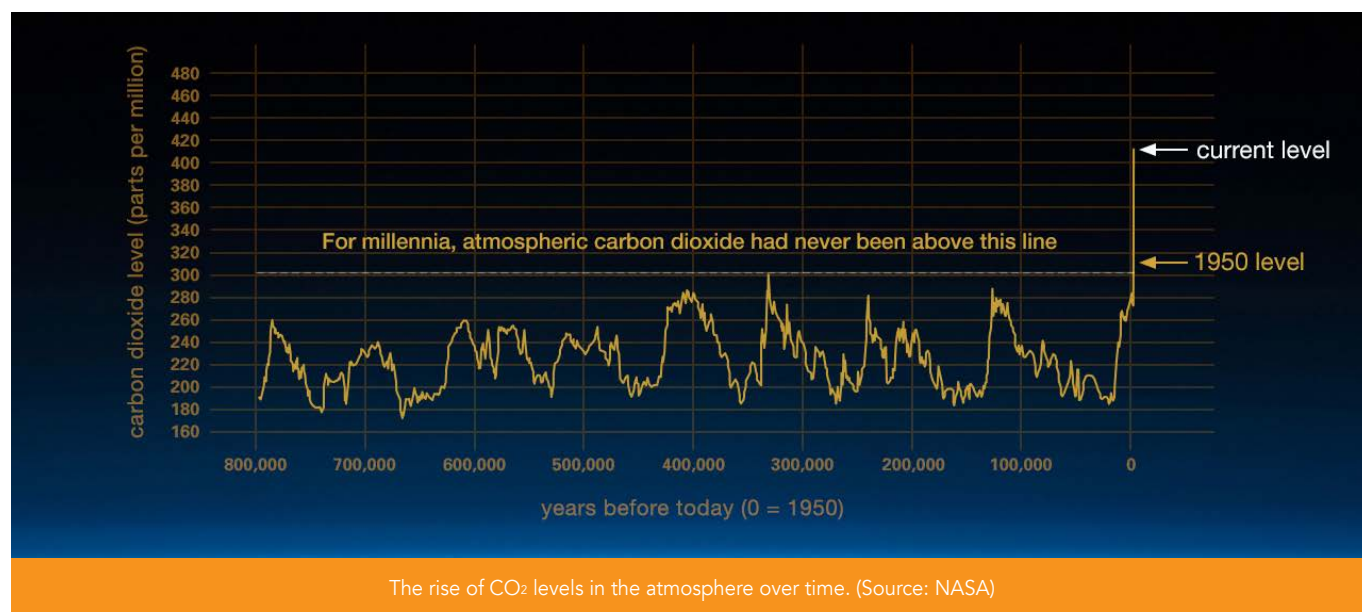
cyclones. Increased temperatures are also linked to increased incidents and severity of wildfires and to heatwaves, which impact negatively on both human health and the environment.

The changes we are already observing include:

- Rising global temperatures;
- Changes in the amount, intensity and frequency of precipitation (rainfall, hail, etc.);
- Increasing number and intensity of extreme weather events (such as droughts, heat waves, floods, tropical cyclones and tornadoes);
- Sea level rise;
- Melting of Arctic ice sheets and mountain glaciers and decreased snow cover;
- Warming of oceans and ocean acidification;
- Changes in the timing of seasonal events such as the arrival of the summer season;
- Ecosystem shifts and biodiversity loss.

In addition to the immediate risks these changes pose, these physical impacts in turn put pressure on our food and water systems, with negative impacts on human health and wellbeing as well as the environment.

How much climate change impacts us in the future depends on how much we do to limit our greenhouse gas emissions and reduce global warming, and how much we do to help people to adapt to the changes that are already inevitable.



^[2] IPCC, 2021

^[3] <https://climate.nasa.gov/evidence/>

WHAT CAN WE DO ABOUT IT?

CLIMATE CHANGE MITIGATION

One of the most important steps we need to take to address climate change is to reduce the quantity of greenhouse gasses humans release into the atmosphere.

This type of action, which includes efforts to both reduce the sources and enhance the absorption of greenhouse gases, is known as climate change mitigation.

Climate change mitigation requires that we change the way we power our societies and use resources, shifting from a reliance of fossil fuels to clean, renewable energy

Projected Climate Impacts in South Africa

We use climate models to help us predict what changes we can expect due to climate change, and downscale global models to give us regional data for South Africa. Although we can project some changes with a fair level of certainty, these models can never be perfectly accurate, and how much change we experience will also depend on how much action the world takes to limit greenhouse gas emissions going forward. The effects of climate change vary by region and negative impacts are not evenly distributed around the world - South Africa is already experiencing warming at about 1.5 times the global average. The following are some of the climate changes we can expect in South Africa:

Temperature

Climate models predict that both the average minimum and maximum temperatures are likely to increase significantly across most of South Africa, and we are likely to experience more extremely hot days. Going forward we can expect temperatures in the interior in particular to continue to rise, with the east coast likely to see greater temperature increases than the west due to the cooling effect of the cold Benguela current. Average temperatures are likely to increase by between 1 and 3°C by 2050 depending on how much and how quickly greenhouse gas emissions are reduced.

Rainfall

Changes in rainfall are more difficult to predict and are unlikely to be uniform across the country or across season, although overall rainfall is more likely to see some decline by 2050. By this time-period there is some indication that there might be increases in winter rainfall over the south-western Cape and east coast and eastern escarpment. Although rainfall projections remain very mixed, there is likely to be a higher frequency of flooding events. Southern Africa has already experienced a drying trend over the western regions in the past 50 years, with increasing numbers of dry spells. This drying trend is likely to continue to 2050, and south-western regions will be at risk of more frequent droughts during the spring and summer months. Rising temperatures combined with drier conditions are likely to significantly increase evaporation rates, affecting the availability of soil moisture.

Sea level rise

The sea level is likely to rise by about 0.13-0.4 meters by 2030, and by as much as 2.5 to 6.5 metres by 2050, with significant impacts during storm surges. This will impact on low lying coastal areas.

Source: LTAS, 2013

sources. Actions we can take include the wide-spread adoption of renewable energy sources like solar and wind energy, improving transport systems to provide reliable public transport and switching vehicles to use renewable fuel sources, and implementing energy efficiency measures to reduce our energy use. At the same time we need to stop deforestation and restore the ability of natural habitats to absorb and store carbon.

Implementing these measures requires both individual action and policy changes from governments around the world. In order to coordinate these efforts at a global level, countries have signed up to the UNFCCC and Paris Agreement treaties (see box text), and have undertaken a range of commitments to reduce their greenhouse gas emissions. The current globally-agreed goal under the Paris Agreement to limit global warming to 1.5°C compared to pre-industrial levels will require a concerted and urgent global effort to achieve. Without these limitations, current trends indicate that we would reach a level of global warming of around 4°C by the end of the century^[4].



The UNFCCC and the Paris Agreement

United Nations Framework Convention on Climate Change (UNFCCC)

Countries from around the world agree on what action they will take to address climate change under the UNFCCC. This convention, established in 1992, has nearly universal membership with 197 members, and is the parent treaty to the Kyoto Protocol and the Paris Agreement. The goal of this convention is to stabilize greenhouse gas concentrations in the atmosphere at a level that will prevent dangerous human interference with the climate system, in a timeframe which allows ecosystems to adapt naturally and enables sustainable development. States that are party to the Convention meet annually at the UNFCCC Conference of the Parties (COP) to review progress and make decisions on how to address climate change, as a global response is required to address climate change.

Paris Agreement

The Paris Agreement is a legally binding international treaty adopted in 2015, which aims to limit global warming to well below 2°C (preferably no more than 1.5°C) compared to pre-industrial levels. This requires that countries limit their greenhouse gas emissions as soon as possible to reach climate neutrality by 2050. In order to achieve this, countries are required to submit plans for climate action known as Nationally Determined Contributions (NDCs), outlining how they will reduce their greenhouse gas emissions and build resilience to adapt to the impacts of climate change. These commitments must be revised every 5 years and go along with commitments from developed countries to provide developing countries with financial, technical and capacity support to address climate change.

⁴ IPCC, 2021

ADAPTATION TO CLIMATE CHANGE

Climate change is already a reality, and countries and communities need to plan for these changes and take a range of practical steps to protect themselves from the disruption and damage to human and natural systems that we anticipate.

To thrive as a society, we must reduce our vulnerability to the harmful effects of climate change and aim to take advantage of any opportunities it may present. Exactly which actions are optimal will vary from place to place. The uncertainty surrounding climate impacts and limited resources combine to make planning and implementing these actions challenging.^[4]

Planning for adaptation therefore requires a process of understanding local risk factors and developing locally appropriate plans to address these. For example, if we can predict that the local climate is likely to be warmer and drier in a region, local farmers may need to diversify the types of crops they plant and adjust their farming methods to cope with these conditions.

At times infrastructure changes may also be needed to cope with impacts such as sea level rise or increasingly intense extreme events such as flooding, or early warning systems may need to be implemented.

Communities face varying levels of vulnerability to climate impacts and have different levels of access to resources to support the process of adaptation - climate change is likely only one of a range of stressors that they face. A key part of this response is therefore to build the resilience of communities and ecosystems to withstand a range of shocks and changes.

Effective adaptation requires concerted efforts across all sectors and at all levels of government, engaging a wide range of stakeholders from local communities to the private sector and civil society organisations.

Although climate change is a global issue, the impacts will be felt at a local level, and local-level action on adaptation will therefore be crucial. If we all take responsibility for our actions in this regard we will be able to make a difference.



Rooibos farmers in the Suid Bokkeveld region. (Source: Noel Oettle, Indigo, Suid Bokkeveld, Northern Cape Province, South Africa)

LINKS FOR MORE INFORMATION

- The Intergovernmental Panel on Climate Change (IPCC) publishes regular reports collating the latest scientific knowledge on climate change. The 5th Assessment report was published in 2014:
<https://www.ipcc.ch/assessment-report/ar5/>
while the 6th Assessment Report (AR6) is being released in 2021/2022 [LO1] [EKL2]:
<https://www.ipcc.ch/report/ar6/wg1/>
- The United Nations Framework Convention on Climate Change: <https://unfccc.int/>
- The Paris Agreement: <https://unfccc.int/process-and-meetings/the-paris-agreement/the-paris-agreement>
- America's National Aeronautics and Space Administration (NASA)'s Global Climate Change information portal: <https://climate.nasa.gov/> and Climate Kids resources: <https://climatekids.nasa.gov/>
- National Geographic's Classroom Resources library on the Greenhouse effect:
<https://www.nationalgeographic.org/encyclopedia/greenhouse-effect/>
- Climate Change Information Fact Sheet: South Africa (USAID, 2015):
https://www.climatelinks.org/sites/default/files/asset/document/South%20Africa%20Climate%20Info%20Fact%20Sheet_FINAL.pdf
- South Africa's Long Term Adaptation Scenarios (LTAS), Summary for Policy-makers:
https://www.environment.gov.za/sites/default/files/docs/summary_policymakers_bookV3.pdf
(Department of Environmental Affairs, 2013)

This introductory series on the impact of climate change on small-scale farmers is published by the ***Innovative Partnerships for Change: Raising New Climate Change Initiatives and Champions*** project.

This project is being implemented by the Association for Rural Advancement (AFRA), Indigo Development and Change, the Institute of Natural Resources (INR), the Farmer Support Group (FSG) of the University KwaZulu-Natal and Heifer Project South Africa (HPSA).

The ***Innovative Partnerships for Change Project*** seeks to support small-scale farmers to effectively adapt to climate change.

The overall objective of the project is to develop a united and cohesive national civil society strategy to address climate change impacts on the land tenure and farming systems of small-scale farmers in South Africa. A key element in achieving this objective is to ensure that land and agriculture sector Civil Society Organisations (CSOs) are capacitated to understand the causes and impacts of climate change.

This series of 12 publications is a resource that seeks to develop and deepen this understanding on the part of CSO's.

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